Chapter 3
Molecular Mechanism of Self-Organization in Biological Systems

Abstract Self-organization is the autonomous formation of complex structures from units of less complexity by local internal interactions. As a precise model of self-organization in nature, embryonic organogenesis is a process where different tissues and organs form in a growing embryo by the autonomous assembling of cells together.

Embryonic self-organization, which occurs at different levels of complexity from nano to macro levels in biological systems, is a highly efficient autonomous process. Self-organization is like robotics without wires and motors. It means the manufacturing program is embedded in materials themselves.

When we look at the manufacturing and construction process through different industries, there are major efficiency issues in energy consumption and labor work compared to autonomous formation and assembling of system's parts during self-organization.

In this chapter, we will discuss the regulatory mechanisms behind embryonic organogenesis through the information storage in biomolecules. In addition, specifically, we discuss on molecular regulation of both differentiation and morphogenesis and their application in regenerative medicine. Deep understanding of regulatory mechanisms of self-organization can open new avenues in designing next generation smart, self-organizing materials and systems for both industrial and biomedical applications.

Keywords Self-organization · Embryonic organogenesis · Morphogenesis · Protein folding · Autonomous manufacturing · Programmable materials · 3D and 4D printing · Cognitive chemistry

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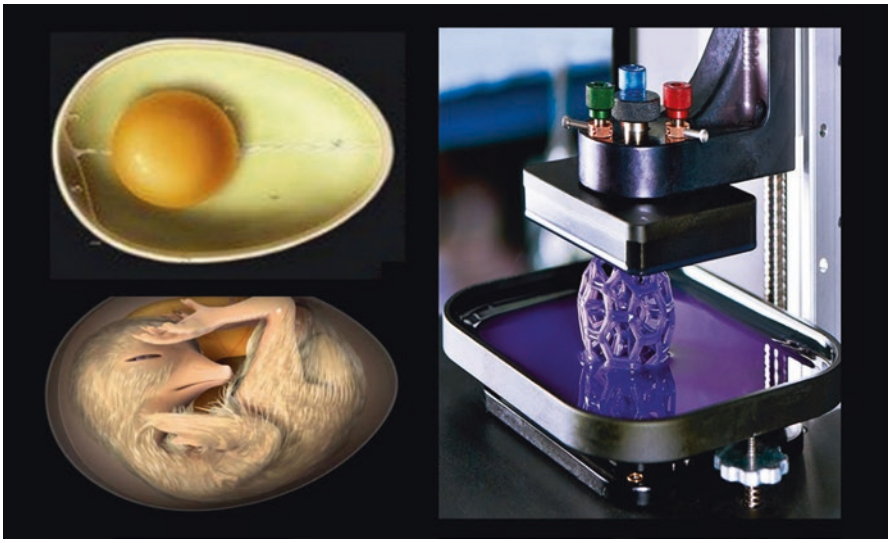

"It is difficult for us in the twenty-first century to realize how totally mysterious embryological development has been" [5].

3.1 An Introduction to Self-Organization and Embryonic Organogenesis

The above cartoon shows the fundamental principles of autonomy in embryonic self-organization (an internally programmed process), compared to the generation of an organized construct by a 3D printer (an externally programmed process).

During embryonic self-organization, a complex and exquisitely ordered body is generated from the relatively unorganized fertilized egg. "It is difficult for us in the twenty-first century to realize and explain all molecular aspects of self-organization process that happens in growing embryo at early stages of development" [5]. Much remains to be elucidated about the relationships among the complex molecular systems of living entities, how they are constrained by the system as a whole as well as by physical laws. "Indeed, it is still an open question for some as to whether we have yet a sufficiently rich understanding of the laws of nature or whether we need to seek deep laws that lead to order and organization" [8–10]. "The fourth law of thermodynamics about such phenomena may ultimately be needed" [10].

Self-organization is the autonomous formation of complex structures from units of less complexity by local internal interactions of systems subunits [13, 19, 26]. Embryonic organogenesis in animals and germination in plants' seeds are precise models of self-organizing systems in nature [26, 31].

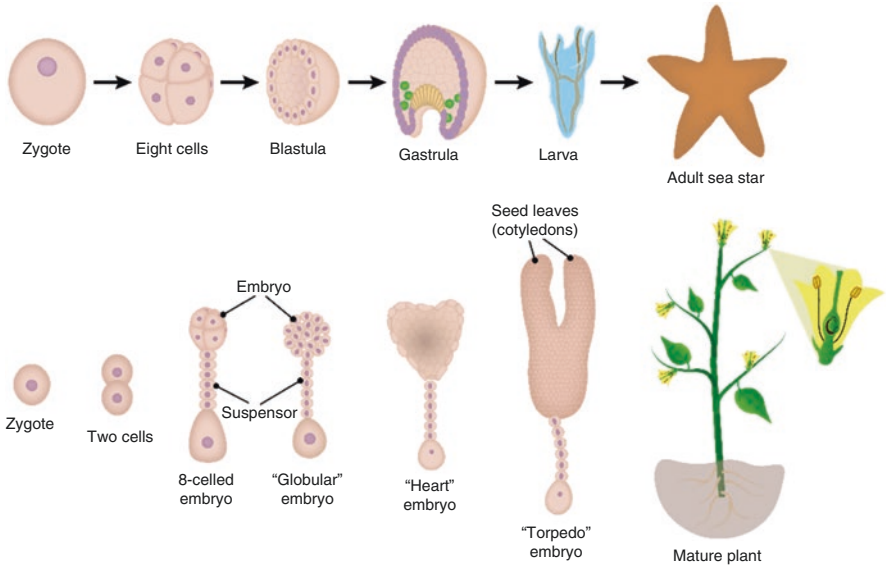

Fig. 3.1 (**a**, **b**) Schematic image representative of examples of self-organization in biological systems. Spontaneous formation of highly organized geometrical structures during the organogenesis in an animal fertilized egg (**a**) and germination of a plant's seed (**b**)

Figure 3.1 shows the fundamental phenomenon of autonomy during self-organization in a fertilized egg and a plant's seed. Organogenesis in animals is a process by which different types of cells, tissues, and organs gradually form at early stages of embryonic development. In plants organogenesis occurs continuously and only stops when the plant dies. For example, formation of lateral leaves, flowers, and branches in the shooting continues constantly during the plant's life time [31]. Organogenesis is an autonomous process which means the instructions and programs for assembling of system's subunits are embedded in materials of the system by themselves [6, 19, 26]. In fact, organogenesis in a living form is a unique autonomous processes which integrates the units of software (instructions and programs for the progress of process) and hardware (materials that constructs the system's physical mass) all together. In the other word, there is no gap between the software and hardware.

Furthermore, beyond the autonomous assembling of programmable materials, embryonic development is accompanied by Computational capacity of materials.

Self-organization is like robotics without wires and motors. It means the manufacturing program is embedded in materials themselves. This is the ability to program physical materials to change shape and physical properties. In addition to the formation of structures, during the embryonic organogenesis, thousands of functional and machinery units form constantly at multiple levels of complexity (starting from molecular to micro and macro scales) at different tissues and organs to operate a collectively uniform system. This collective connection between the molecular subunits of a living organism functions as molecular algorithm which runs all developmental operations during the embryonic organogenesis.

3.2 Potential Applications of Self-Organization in Technology and Biomedical Sciences

Understanding self-organization phenomena could lead to several breakthrough applications and improvement to existing technology. Some of the potential use cases are listed below further applications of self-organization in biomedical science will be discussed in Chap. 6.

- Generation of synthetic self-organizing systems
- Generation of synthetic self-organizing material as an alternative approach for 3D printing
- Generation of self-healing systems
- Stem cell-based regenerative medicine and tissue engineering
- Treatment of protein misfolding diseases

Unlike self-assembly that can be observed in both living and non-living systems; self-organization is a specific property of life and can be observed only in living organisms. In fact, there are several features in self-organizing systems which distinguish them from self-assembly, including structural complexity and energy balance of the system. In the next section, we will discuss the main properties of self-organization and how it can be distinguished from self-assembly.

3.3 Remarkable Properties of Self-Organization

3.3.1 Polarity Formation

One of the most remarkable properties of self-organization is the capacity of polarity formation, which also can be defined as the symmetry breaking capacity of the system [20].

Polarity formation involves autonomous generation and assembling of highly specific geometrical structures, which can grow asymmetrically through different layers of complexity from nanostructures to micro- and macrostructures (Figs. 3.1 and 3.2) [13]. Therefore, self-organization is accompanied by the formation of geometrical structures with highly specific patterns and morphologies. The process also is called morphogenesis.

Figure 3.1 illustrate examples of polarity formation in growing animal and plant embryos. Figure 3.2 compares the special features of self-organization (including the polarity formation and generation of multilayer structural complexity) in a plant versus the uniform molecular organization of self-assembling crystals. Figure 3.3 illustrates the autonomous reorientation of cells and establishment of body axes in a growing amphibian embryo [2, 11]. Polarity formation in a self-organizing system requires a precise internal program and a source of energy to guide and enforce the progress of the process. Thus, self-organization only occurs through a dedicated internally programmed process [27, 31].

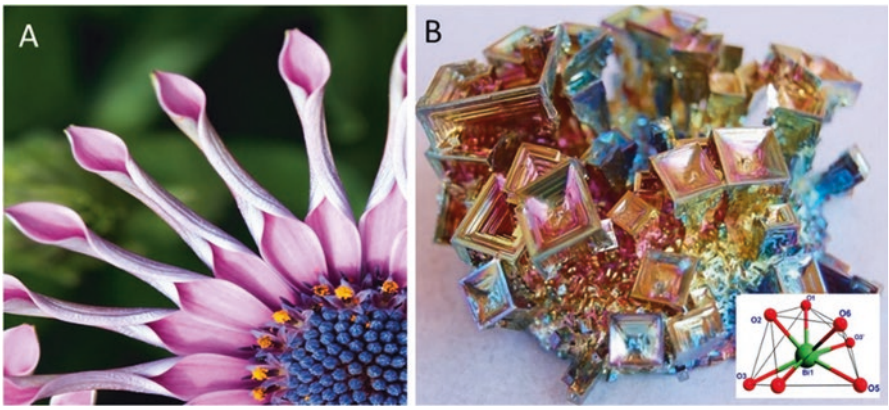

Fig. 3.2 (**a**, **b**) Comparison of self-organization and self-assembly. Polarity formation and multi-layer structural complexity formation in a self-organized plant (**a**) compared to the uniform molecular structure of a self-assembling crystal (bismuth crystal)

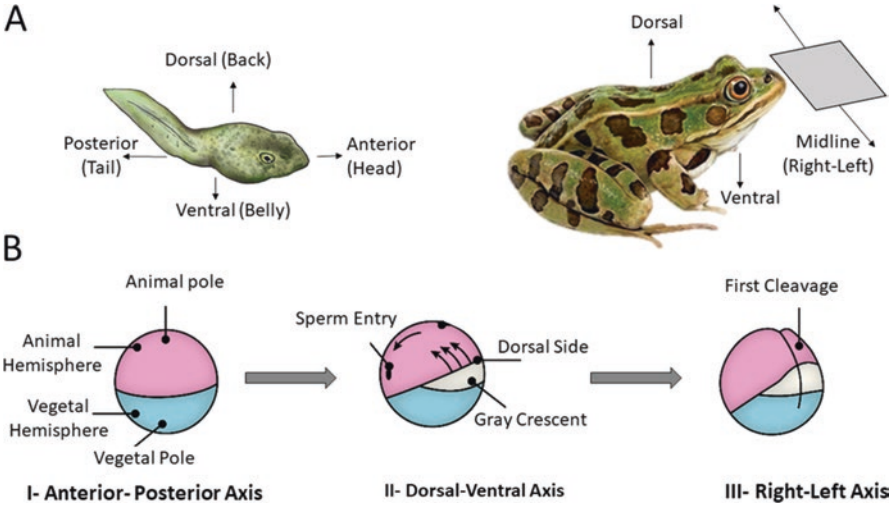

Fig. 3.3 (**a**, **b**) Gradual polarity formation and generation of body axis in a growing amphibian embryo. (**a**) Schematic image illustrates the three axes over the body of a developed embryo. (**b**) Schematic image illustrates the gradual polarity formation, reorientation of cells, and establishment of body axes during the early stages of embryonic development

3.3.2 Energy Balance

Unlike self-assembly that can be observed in both living and nonliving systems (e.g. formation of ice crystals), self-organization can be observed only in living systems. Self-assembly can be explained by thermodynamics and through the stability of the

molecular structure [1]. However, the formation and maintenance of a self-organizing system rely on continuous energy input into the system. Thus, by considering the energy balance of the system, self-organization requires the autonomous energy uptake from its environment. As another remarkable natural property, biological systems are self-fueling and can autonomous manage their energy requirements [20]. Mechanisms of energy management and self-fueling properties of biological systems are discussed in Chap. 4 in detail.

3.4 Self-Organization Is a Bottom-Up Process

Self-organization is a bottom-up process which starts from the molecular level and progresses toward micro and macro levels of complexity. Protein folding is a great example of self-organization at the molecular level. Protein folding is a process in which molecules of proteins obtain their 3D structure spontaneously [16, 17, 28]. In fact, proteins do not need an external scaffold to obtain their complex structure; instruction for the 3D structure of proteins is embedded in the sequences of amino acids. Specifically, chemical codes are embedded in the carefully ordered electrochemical interactions between chemical groups in the side chains of amino acids which lead the spontaneous formation of a unique 3D structure for each molecule of protein. This is the most stable condition compared to all other folding possibilities (Fig. 3.4). Figure 3.5 illustrates the spontaneous formation of the 3D structure of insulin molecule which is a repeatable process [18, 23].

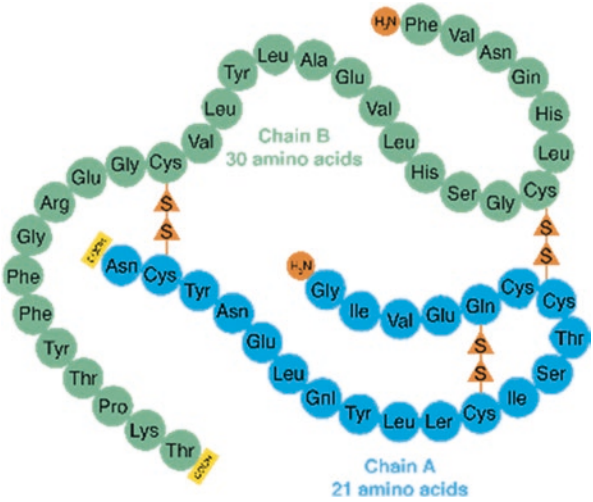

Fig. 3.4 Illustration of the spontaneous folding of insulin molecule and formation of its unique 3D structures

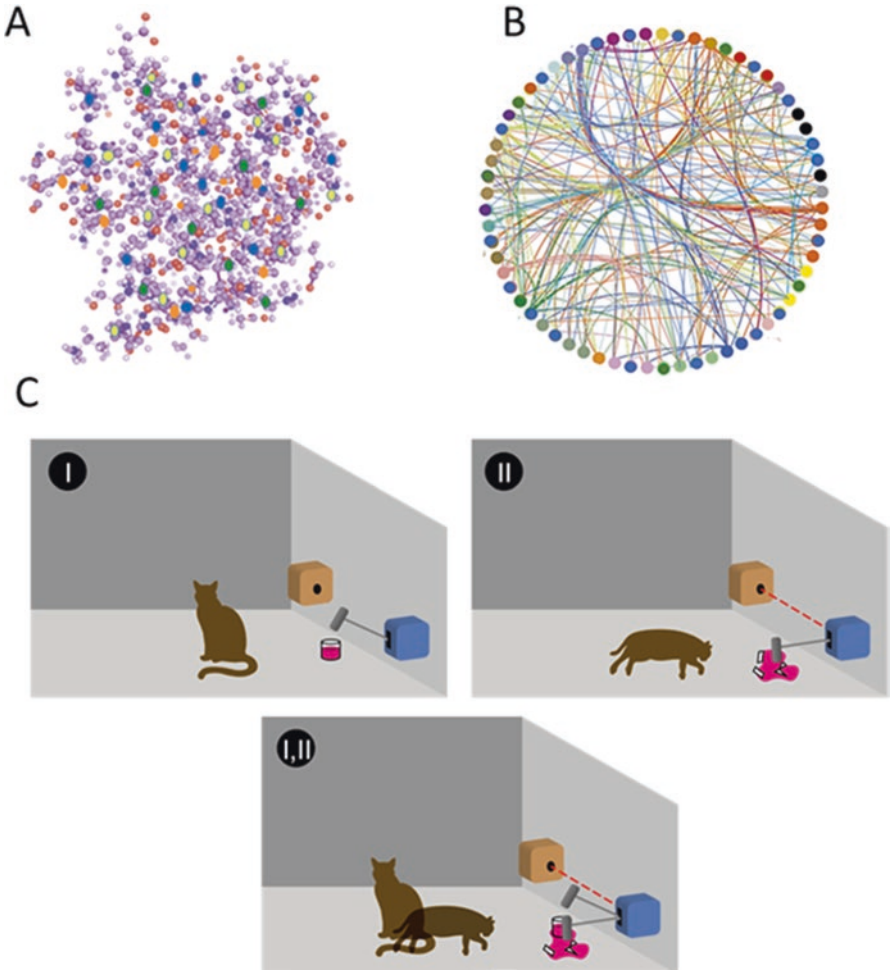

Fig. 3.5 (**a**–**c**) The schematic image illustrates the efficiency of autonomous and repeatable protein folding through electrochemical attractive forces among atoms (**a**, **b**). Real time, spontaneity, and repeatability of protein folding can be compared with quantum theory; meaning despite the existence of all different types of molecular interaction and large number of possible senarioes during protein folding, only one condition happens in real time which is the most stable structure

As previously mentioned, self-organization is a bottom-up process which progresses through multiple layers including nano, micro, and macro levels of complexity [11]. Figure 3.6 provides an example of self-organization at the molecular level in a protein molecule.

Figure 3.7 refers to the progress of self-organization at the levels of cells, tissues, organs, and the entire body of an organism. At the cellular level, during organogenesis, embryonic stem cells are able to assemble together to make all complex structures in different tissues and organs.

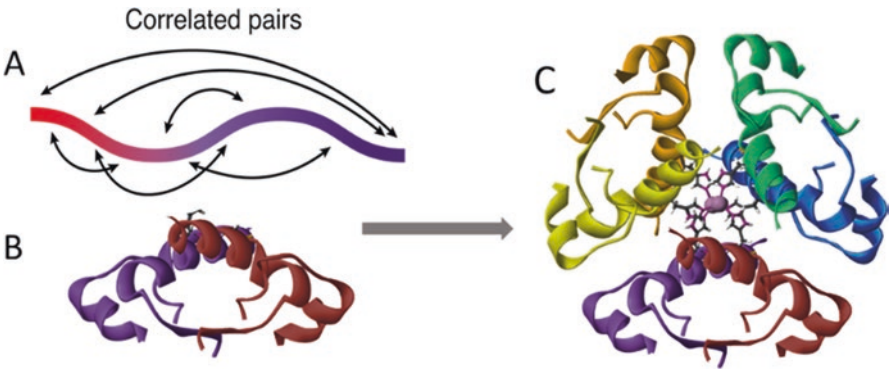

Fig. 3.6 Illustrates an schematic image representative of sequences of events (A-C) during protein folding. Proteins are able to obtain their conformational structure by the information that is instructed in sequences of their amino acids

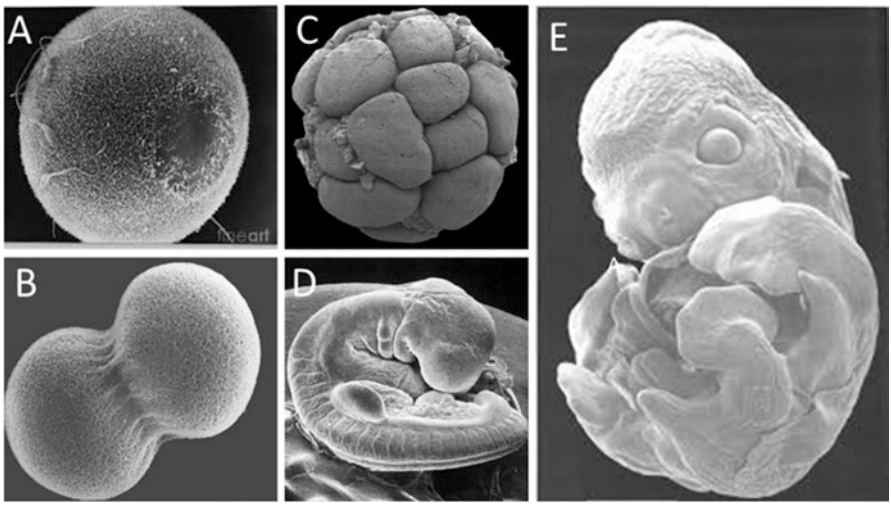

Fig. 3.7 Schematic image representative of self-organization of stem cells during the early stages of embryonic development. Stem cells are able to assemble together and make highly specific structures, which grow in size and complexity by time. Scanning micrographs of the human embryo at different stages (A to E) of development (Carnegie stages). (Images are reproduced with the permission of Dr. Kathy Sulik, from collection of Dr. Vekemans and Tania Attie-Bitach. Source of images: https://embryology.med.unsw.edu.au/embryology/index.php/Carnegie_stage_table)

When we study embryonic organogenesis through multiple layers of complexity, we realize that more than structure formation, self-organization in biological systems is accompanied with more complex functions, including individual and collective cell migrations towards specific destinations, polarity formation and continues diversity formation in the primary population of cells.

Unlike self-assembly which can be explained simply by electrochemical interactions of molecules through thermodynamics, operation of more complex functions of self-organization (e.g. cell mitigation, and polarity formation) requires precise internal programs to guide the system. As we discussed in chapter 2, all cell functions are operated by molecular circuits.

Most of computational functions of cells are operated by protein signaling networks. In fact, proteins can execute nearly all operations in a cell.

Thus, enzymes provide the intricate molecular surfaces which function as microchips in a cell. Other proteins function as sensors, recognition modules and signal transducers. In addition, 4D structural flexibility of proteins makes their logic gate functionality, and provides a unique capacity to respond to the environmental factors, directly. This leads to a very high computational capacity for proteins as individually operational units and in connection with other proteins and signaling molecules in dynamic cell signaling networks.

In the next part we discuss the role of cell molecular regulation behind two critical processes that occurs during embryonic self-organization, including, 1- pattern formation and and 2- cellular specification and differentiation.

3.5 Developmental Biology of Embryonic Organogenesis

3.5.1 Developmental Biology of Autonomous Morphogenesis and Spontaneous Pattern Formation

Morphogenesis is originated from two Greek names including morphe (means, shape) and Genesis (creation). It is a process that causes an organism to develop its shape. The process controls the organized spatial distribution of cells during the embryonic development of an organism. More than additive assembling of structural subunits, embryonic organogenesis is accompanied by more complex behaviors and functions including, 1- individual and collective mitigation of cells towards special destinations 2- commitment, specification and differentiation of initial stem cells towards different cell lineages. Morphogenetic responses may be induced in organisms by hormones, environmental chemical factors including substances produced by other cells, Signaling molecules with the maternal origin, or by mechanical stresses induced by spatial patterning of the neighboring cells. Furthermore, morphogenesis is accompanied by polarity formation in the conformational shape of a growing embryo [5].

Formation of general polarity in the embryo is instructed by the embryo's genome. However, the egg cytoplasm also plays a critical role in determining the patterns of cleavage and cell specification at the early developmental stages. Once fertilization is complete, the development of a multicellular organism continues by a process called cleavage, a series of mitotic division whereby the enormous volume of egg cytoplasm is divided into a large number of smaller cells. In fact, fertilization

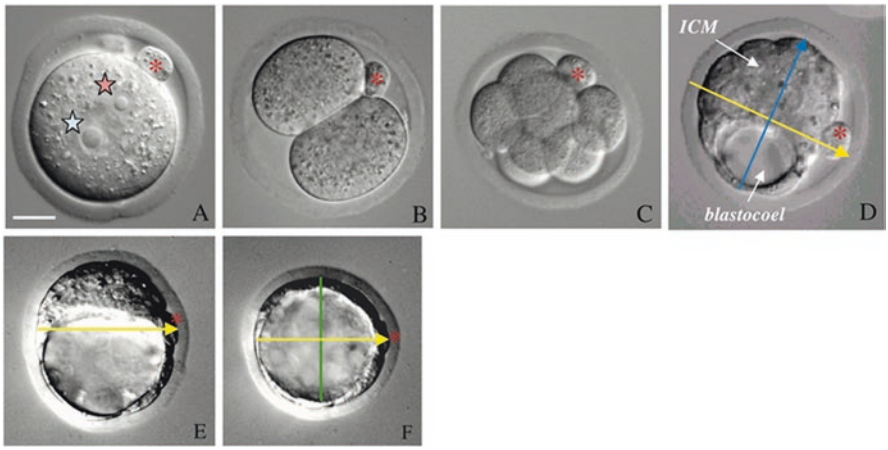

Fig. 3.8 (**a**–**f**) Cleavage and polarity and axis formation at early stages of embryonic development (Image adapted from Zernicka-Goetz [32])

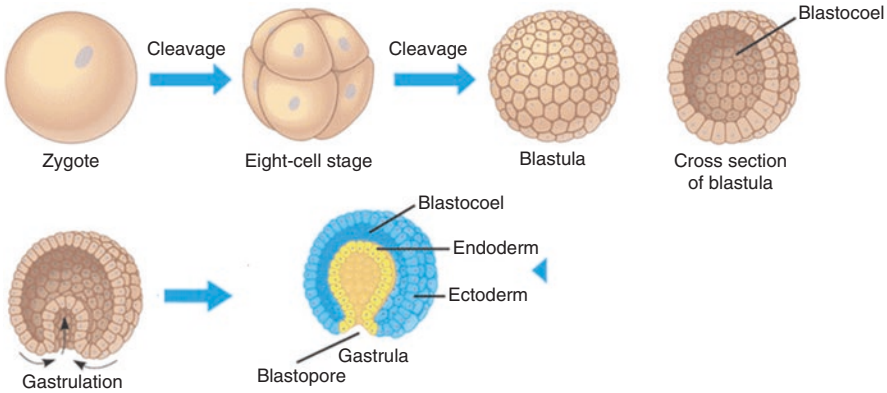

Fig. 3.9 Schematic image representative of movement of cells during gastrulation and gradual polarity formation from the nonpolar structure of blastula toward the polar structure of gastrula [5, 24]

gave the organism a new genome and rearrangement of its cytoplasm. These cleavage-stage cells are called blastomeres [5, 13] (Fig. 3.8).

Fast-dividing embryonic cells then later undergo spontaneous displacements and begin to move and migrate toward different parts of the embryo and acquire new neighbors. The displacement and migration of dividing embryonic cells lead the formation of early embryonic polar structures including the blastula and gastrula (Figs. 3.8 and 3.9). The blastula consists of numerous embryonic stem cells, the positions of which were established during cleavage [21].

In the next stage which is gastrulation, multilayered body plan of organism forms. Migration and rearrangement of cells during gastrulation causes the formation of three germ layers including ectoderm, mesoderm, and endoderm (from the lateral to the medial part of embryo's body) (Fig. 3.11).

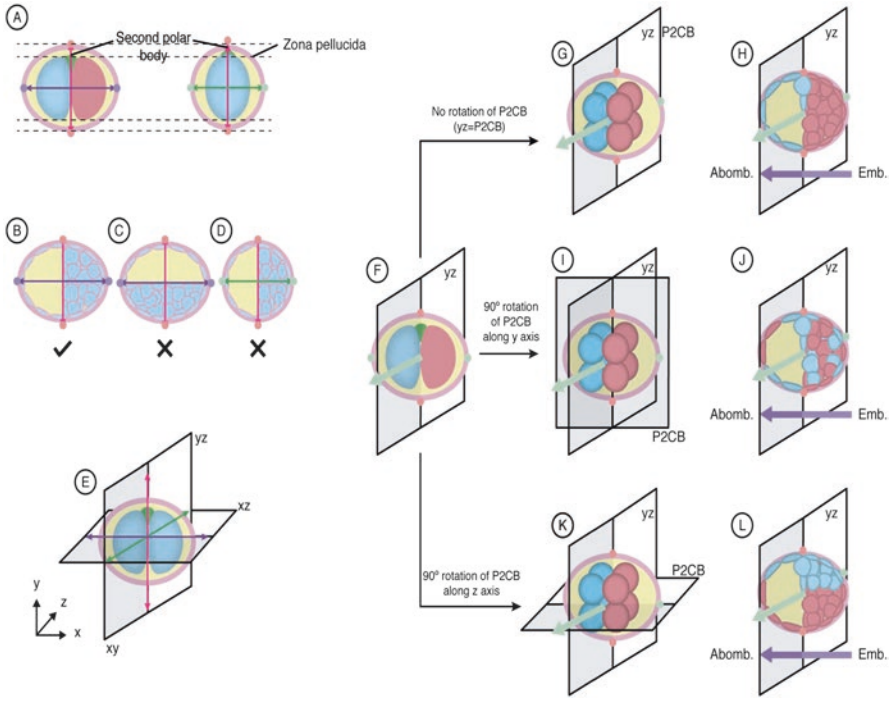

Fig. 3.10 (**a**–**k**) Schematic image representative of the axis by collective movement (migration) of embryonic stem cells and formation of blastula. (Image is adapted from Rossant and Tam [24])

The cells that will form endodermal and mesodermal organs are brought inside of the embryo, while the cells that will form the skin and nervous system are spread over its outside surface (Fig. 3.11). Thus, the three germ layers – outer ectoderm, inner endoderm, and interstitial mesoderm – are first produced during gastrulation. In addition, the stage is set for the interactions of these newly positioned tissues [3, 5, 21] (Fig. 3.11).

Based on results of studies on embryonic development, we know that different organisms undergo cleavage in distinctly different ways. While cleavage always precedes gastrulation, axis formation in some species can begin as early as oocyte formation (Fig. 3.10). The pattern of embryonic cleavage peculiar to a species is determined by two major parameters: (1) the amount and distribution of yolk protein within the cytoplasm and (2) factors in the egg cytoplasm that influence the angle of the mitotic spindle and the timing of its formation. During cleavage and gastrulation, the major axes of the embryo are determined and the embryonic cells begin to acquire their respective fates [5, 21] (Figs. 3.10 and 3.11).

The challenging question is: what is the regulatory mechanism behind polarity and pattern formation during and after gastrulation? Which factors are regulating the single and collective migration of embryonic cells toward the formation of specific morphologies?

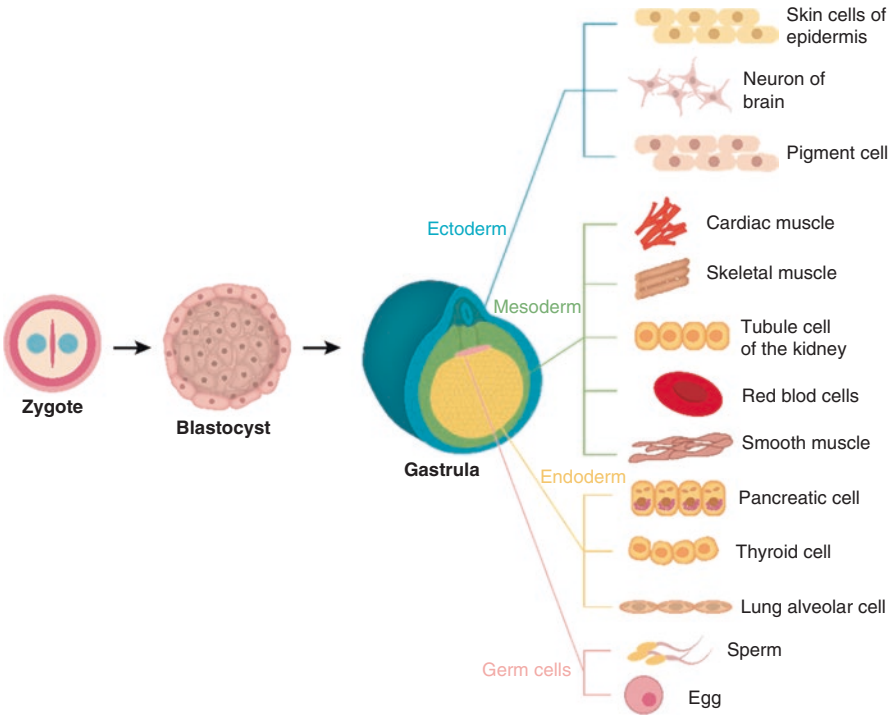

Fig. 3.11 Schematic image representative of formation of three germination layers as well as differentiation into all different cell types in the body

Cell fates are specified by an asymmetric distribution of patterning molecules. These patterning molecules also are called morphogens. A morphogen is a biomolecule that causes the fate determination of a cell by its concentration gradient.

Morphogens can be transcription factors produced within cells. They can also be paracrine factors that are produced in one group of cells and then travel to another population of cells. Various combination and concentration of morphogens can produce different biological signals for developing stem cells. Concentration gradients of morphogens produce different signals for the exposed cells, which cause the specification of the target cells [25] (Fig. 3.12). Furthermore, concentration gradient of morphogens causes the polarity formation at different parts of embryo's body.

Morphogenesis as an old mystery in the history of science, in addition to the biologists, has drawn the attention of researchers in other fields of science. Alan Turing's accomplishments in computer science are well known, but lesser known is his research interests in understanding the algorithms behind the spontaneous morphogenesis in biological systems. In his only published paper on biology, Turing proposed a mathematical model of morphogenesis. Turing offered a theory for morphogenesis based on the chemistry and through a process called intercellular reaction-diffusion [29].

In this model, a set of chemicals react with each other and diffuse across space, between cells in an embryo [5, 29, 30]. These chemical reactions need an inhibitory

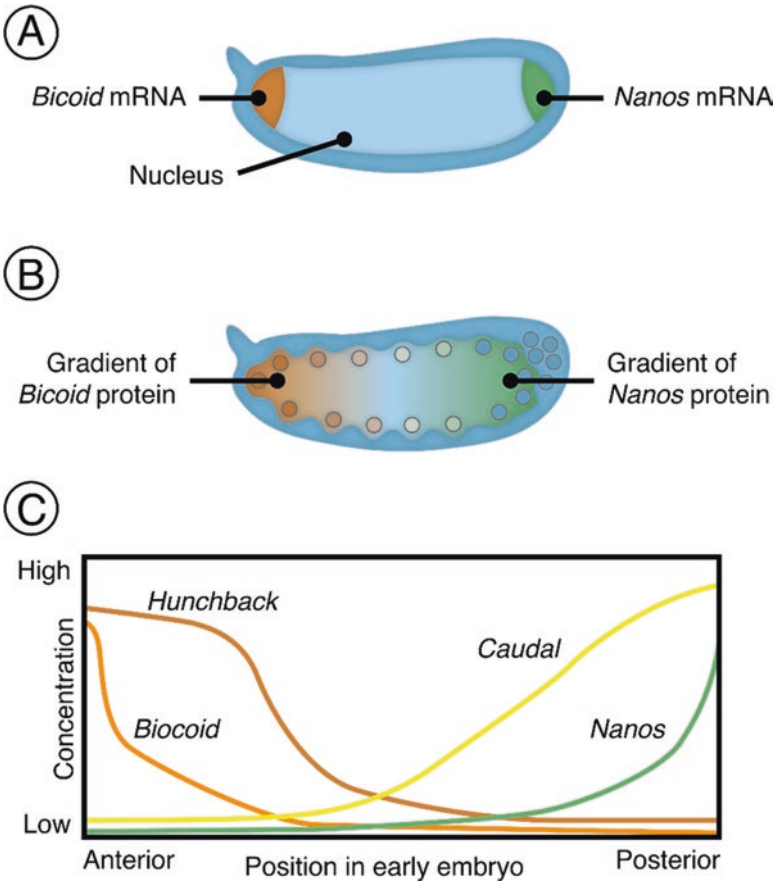

Fig. 3.12 Gradients of different morphogens in *Drosophila* embryos. Bicoid and Hunchback, regulate the production of anterior structures, while, combinations of Nanos and Caudal regulate the formation of the posterior parts of the embryo [2, 5, 11, 14]

agent to suppress the reaction and an excitatory agent to activate the reaction (Fig. 3.12 explains the mechanism of morphogenesis by diffusion chemical gradients) [5, 14, 29]. The accuracy of intercellular reaction-diffusion theory later was proved by several biologists through the polarity formation effect of chemical gradients [12, 14, 25]. Figure 3.12 is a schematic image that illustrates the mechanism of polarity formation at early stages of embryonic development in *Drosophila* [5, 14].

The formation of anterior-posterior polarity of the body is instructed by the anterior-posterior distribution of morphogens in the egg. The initial morphogens may originate from mother's ovaries. Gradients of maternal mRNAs encode transcriptional and translational regulatory proteins that diffuse through the syncytial blastoderm and activate or repress the expression of certain zygotic genes. Appropriate gradient concentrations of Bicoid and Hunchback, regulate the production of anterior structures, while, Nanos and Caudal, regulates the formation of the posterior parts of the embryo [2, 5, 11, 14].

In summary gradient concentration morphogens leads to the dynamic rearrangement of cells which can be observed through the individual and collective cell migrations. Further, we should keep in mind that more than physical rearrangement of cells, during the morphogenesis, individual cells also undergo internal biochemical changes which leads the formation of structural and functional diversity in the initial stem cells' population.

These changes in cellular biochemistry and function are operated by reciprocal interactions between the inter and extra cellular molecular circuits. AS discussed earlier most of computational functions of cells are operated by protein signaling circuits.

In the next part we refer to the main protein signaling pathways that are involved in specification and differentiation of stem cells and formation of embryonic germ layers.

3.5.2 Developmental Biology of Autonomous Specification and Differentiation

As discussed above, morphogen gradients instruct the specific arrangement of stem cells and pattern formation during the early stages of embryonic development. After the fertilization, cell growth starts with a homogeneous population but gradually shifts to the formation of heterogeneous cells which can be distinguished from each other by biochemical and physical properties. Differentiation is only the last overt stage of a series of events that commit the developing population of stem cells to become a particular cell type. For example, a skin cell obviously differs in its biochemical composition and cell structure from a hepatocyte or a cardiomyocyte [15, 21, 27].

The process of commitment of stem cells can be divided into two stages including specification and determination. The specification is a process where stem cells are able to be specified autonomously into different lineages when they are placed in a neutral environment such as a petri dish. At this stage, differentiation potential of stem cells is reversible.

The second stage is determination. A cell or tissue possesses the differentiation potential, even when placed into a non-neutral environment (e.g., in another region of the embryo). At this stage, differentiation potential of stem cells is nonreversible [5]. Specification and differentiation of stem cell during germination lead to the generation of three germ layers. The further commitment of cells at each germ layer causes the formation of all different types and tissues in the body (Fig. 3.12). Specification, differentiation, and commitment of stem cells occur through the activation of highly specific cell signaling pathways toward each cell lineage [5].

In addition to the internal program of cell signaling pathways, stem cells' behavior is affected by environmental factors. In fact, stem cells can understand and adapt themselves to the environmental conditions [15, 21, 27].

This means the internal molecular networks are flexible enough to be adapted with dynamic extrinsic signaling stimuli, as well. In fact, stem cells have a unique endogenous cognitive and decision-maker coding system that is able to guide stem cells towards different destinations based on the dynamic interactions with environmental factors.

Below, we will discuss the main genetic and protein signaling networks that regulate early stages of cell lineage specification (from stemness and pluripotency toward the commitment into three germ layers).

Pluripotency is a distinguishing characteristics of stem cells which provides the their differentiation potential towards three main embryonic germ layers. Results of in vitro and in vivo studies on initial population of embryonic stem cells reveals that Oct-4, Sox-2, and Nanog are the key regulatory factors of pluripotency circuitry [21]. Oct-4 is an essential gene regulatory factor which is expressed at the pre-implantation stage. Oct-4 deficient embryos can survive until the morula stage, but, cannot enter to the blastocyst stage in both in vitro and in vivo conditions. Oct-4 binds to the octamer motif (5′-ATGCAAAT-3′) of DNA to control the expression of a cluster of genes involved in pluripotency. For instance, there is a close correlation between the expression of Oct-4 and Sox-2 [27].

Sox-2 belongs to Sox family of transcription factors that have a highly conserved DNA-binding domain. Sox2 expression is widely distributed in different tissues of a developing embryo, including ICM, epiblast, neural tissues, and extra-embryonic ectoderm [21]. Sox-2, is an essential factor for ESC pluripotency, as knockdown or conditional deletion of Sox-2 results in trophoblast differentiation. This phenotype is similar to that caused by Oct4 deletion because Sox-2 often acts as a heterodimer with Oct-4 to regulate transcription of important down-stream genes such as Fgf-4, Nanog, Lefty1.

Nanog is a homodomain-containing protein which functions in coordination with Oct-4 and Sox-2. Genome-wide mapping of Nanog binding sites has identified many pluripotency genes, including Esrrb, Rif1, Foxd-3 and REST [21]. Importantly, many of the self-renewal and pluripotency factors work in cooperation with each other to maintain pluripotency. For example, Sall-4 and Esrrb have been shown to interact with Nanog physically and co-occupy Nanog genomic sites in mouse ESCs [5, 21, 26]. Results of developmental biology studies shows that SMAD1 and STAT3-binding sites are associated with the Oct-4, Sox-2, and Nanog core factor-binding loci. They also share many common regulatory coordinators such as Klf-4, Esrrb, c-myc.

Given that mouse ESCs can be maintained under LIF/BMP condition that enables SMAD1 and STAT-3 activation and binding to genomic sites, this observation provided direct evidence that LIF/BMP signaling supports self-renewal by strengthening core pluripotency circuitry [5, 21]. In the mammalian embryo, the transition from blastocyst to gastrula is a remarkably elaborate process involving a complex series of molecular and cellular events. In addition, expression of Oct-4 and Sox-2 together regulates the cell proliferation though the activation of ERK/MAPK signaling pathway. Furthermore, during preimplantation, successive cleavages of the zygote lead to polarity formation in the body of growing embryo. At this stage

blastocyst cavity is surrounded by the trophectoderm (TE). Subsequently, the primitive endoderm (PrE) becomes apparent at the surface of the ICM in contact with the blastocyst cavity.

Transition from pluripotency to lineage specification occur smoothly by the mechanism of gradient concentration and different combinatorial arrangement of pluripotency signaling factors with the spatiotemporal environmental factors. This leads the activation of downstream lineage specification pathways. For example, high expression level of Sox-2 and Sox-1 is associated with neuroectoderm specification. While high levels of Oct-4 and Nanog along with the up-regulation of Brachyury leads to the mesoderm lineage specification. Wnt/ b-catenin pathway, also play a central role in differentiation of ESCs towards mesoderm lineage [5]. Lineage specification markers such as Fgf-5 and Cer-1 are expressed in the epiblast, but are absent in the ICM [5, 21]. In parallel with activation of lineage specification and differentiation pathways, developing stem cells, gradually change their position and communicate with new neighbors which leads the establishment of multilayer body plans. Collective migration and rearrangement of cells during the gastrulation lead the formation of three germ layers including ectoderm, mesoderm and endoderm and eventually formation of all different tissues and cell types in the organism's body [5, 7, 21–23, 26].

Here our major goal has been to specifically discuss the mechanism of diversity formation during organogenesis and how combinatorial arrangement of genetic and protein networks leads the diversity formation from the initially homologous population of embryonic stem cells. Detailed description of the whole embryonic developmental pathways towards the end stage differentiation of various tissues has been out of scope of this chapter. For detailed information, it is highly recommend reviewing relevant citations in list of references [5, 21, 24, 26, 27, 30].

In overall, we can summarize the main properties of embryonic organogenesis as following:

1. Self-organization is a bottom-up process which starts from the molecular scale and can progress freely through multiple levels of complexity.
2. The process is preciously programmed by an internal molecular algorithm which is also responsive to the physiochemical environmental factors (e.g. through the electrochemical attractive forces between molecules and atoms).
3. The reciprocal interactions of intrinsic cell signaling networks with each other and environmental factors makes a very dynamic operating system for the organism.
4. The process is operated through a dynamic and autonomous algorithm which the out-put of each step functions as an in-put for the next stage.
5. The process is accompanied by autonomous and progressive pattern formation.

By considering the above mentioned properties of self-organization, it can be concluded; Self-organization in biological systems functions similar to a Mandelbrot algorithm, where the out-put of each stage is an in-put for the next stage. Therefore, as long as the operation is running, the algorithm can generates more sequences of dynamic patterns. Also, the algorithm is able to transfer freely among multiple layers of complexity.

Considering the molecular-based property of biological operating systems, it can be concluded; a critical requirement of self-organization is related to the unique capacity of information storage and processing by biomolecules. The tri-fold functional property of molecules also can be translated as relativity in code capacity, and mass for detailed description of different aspects of self-organization. Relativity of code, energy, and mass is a distinguishing property of life, which differentiates a living system, from any other non-living elements of nature.

3.5.3 Molecular and Physical Foundation of Embryonic Self-Organization by Cognitive Chemistry and Relativity of Code Energy and Mass

Can we simulate the developmental mechanisms of self-organization beyond biology?

Scientifically, by considering the chemical foundation of biological systems, it is possible to simulate the self-organizing property of embryonic stem cells. However, the level of complexity of such systems depends on deep understanding the molecular operation system of a living organism and how basic principles of physics and chemistry are applied in designing of living system by nature. Same basic principles can be applied in generation of similar synthetic systems. In the next part of this chapter and also, chapter 6, we try to explain the chemical foundation of self-organization through the basic principles of cognitive chemistry. We will explain properties of optimal materials for generation of self-organizing systems in synthetic forms.

Chemical foundation and molecular properties of self-organizing systems (major characteristics of optimal materials for generation of synthetic self-organizing systems):

1. Internally coded materials and molecules: instruction for the final 3D conformation of a geometrical structure needs to be embedded in the electrochemical attractive forces between atoms of a molecule.

Figure 3.13a illustrates is a schematic image representative of information stored in the sequence of amino acids. Figure 3.13b illustrates the flexible, structure of protein conformation. Unlike any other crystals in nature (with static 3D structure), proteins have highly flexible 3D structures and based on the environmental conditions can obtain different conformational structure. In other words, proteins have a multidimensional geometrical structure in response to the environmental factors. This unique chemical property of proteins leads to their multifunctional properties including molecular signal transducers, logic gate switches, molecular machinery, molecular sensors, etc.

2. High flexibility in the 3D structure of proteins can be defined as the main molecular mechanism of recognition, memory formation, and adaptation with environmental factors in living cells. Dynamic changes in the conformational structure

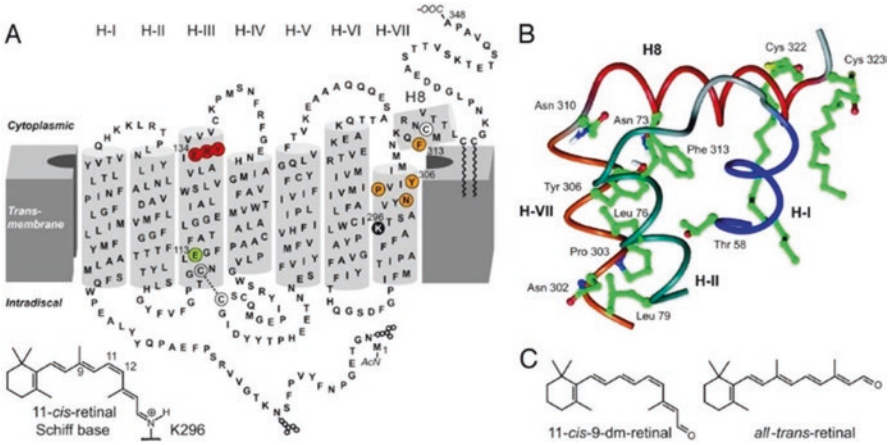

Fig. 3.13 (**a**–**c**) The schematic image represents the multifunctional capacity of proteins (coding units, structural and machinery functions, at the same time). The schematic represents the crystallography of rhodopsin structure. The instruction for the 3D structure of the protein is embedded in the sequences of amino acids, which also is critical in its machinery functionality. (Image adapted from Fritze et al. [4])

of proteins can be observed in molecular sensors, receptors, molecular logic gates, and signal transducers based on environmental factors. Combinatorial interactions of these molecules generate different cell-signaling pathways.

DNA has been the major focus of studies in designing biomimetic coding systems. However, proteins also are promising models for the generation of synthetic coded materials. Proteins have multifunctional capacity due to their special structural properties and are more efficient models for environmental recognition and dynamic memory formation.

Furthermore, proteins are the best models for the generation of self-organizing polymers. All instructions related to the final 3D structure of a protein are embedded in the electrochemical interactions among the side chains of amino acids' sequences. In fact, electrochemical interactions among the side chains of amino acids generate a coding system which regulates the dynamic conformational structure of a protein based on the variations in the environmental conditions. Therefore, the coding system of amino acids instructs the dynamic or 4D structure of a protein, based on environmental factors (signals).

Figure 3.13 illustrates a schematic image representation of the flexibility in the conformational structure of rhodopsin, which is a light-sensitive receptor protein involved in visual phototransduction. Rhodopsin belongs to the category of G-protein-coupled receptors (GPCR) and is found in the rods of retina.

Rhodopsin is extremely sensitive to the light. The 3D conformational structure of rhodopsin changes immediately after exposure to the light. Dynamic changes in the conformational structure of rhodopsin consequently lead to the variation in the affinity of this protein to the other intercellular signaling molecules [4]. This process finally is identified as a biological signal transduction. In fact, unlike any other nonliving crystals (with 3D structure) in nature, proteins illustrate 4D conformational structure.

This unique chemical-based coding system, which is embedded in the sequences of amino acids, is a remarkable property that distinguishes the 4D and dynamic structure of proteins from any other static crystals in nature. The unique coding system of amino acid sequences and the 4D (dynamic) conformational structure of proteins make them function as highly efficient signal transducers and logic gates at the same time.

In fact, each protein functions as a separate computer that is able to store information, not only in the sequences of their subunits but also in their 3D structure. Protein signal transducers are able to convert different types of physicochemical environmental signals into the computable signals for the cells. The multifunctional property of proteins causes a very fast reaction with the environmental signals.

3. In addition, data processing in signaling protein is based on the spontaneous electrochemical interaction of atoms in the amino acids' side chains. Therefore, solving complex problems in the interface with the environmental factors (e.g., NP problems) occurs in real time.

Another example of information stored in the 4D patterns of proteins can be explained by the mechanisms of memory formation in the patterns of brain neural networks and through inter- and intracellular protein and electrochemical signaling pathways among neural cells. Information storage in the 4D patterning of neural cells and protein-protein signaling pathways causes the expansion of memory capacity of neural cells. On the other hand, 4D patterning of proteins during embryonic development is accompanied by cell cleavage that causes the physical growth of developing an embryo.

It is possible that during the activation of neural cells, combinations of different types of neurotransmitters influence the structures of signaling proteins in neural cells. This leads to the formation of completely new structures and activation of completely new sets of signaling pathways. Addition of these pathways causes the expansion of memory capacity of neural cells. Therefore, unlike any other human-made computers with limited information storage capacity, neural cells have expandable memory capacity, meaning the brain memory capacity is expandable as the input information increases (e.g., the brain's information storage capacity enhanced by education).

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